

Radar Search Pipeline TO BE: 0.7.6.3.6.9

Review design for a planned Radar search pipeline change

Slice And Decision Context

Slice 0.7.6.3.6.9: External recall-budget lane reservation and guaranteed expansion execution.

The previous slice fixed target selection before scheduler admission. Docker smoke radar-run-e8936402-b242-4d63-a076-7d563441b7b0 proved that the selector now selects enough guaranteed work:

- holding/group: selected 2, required 1;
- legal/subsidiary: selected 4, required 2;
- production-site/branch: selected 4, required 2.

The run still missed the full production-site guarantee because only one production-site task executed. The immediate cause was exhaustion of openrouter_recall_expansion before all accepted guaranteed tasks had made their first provider call.

AS IS Problem Statement

The scheduler admits guaranteed expansion work, but the external OpenRouter recall-expansion budget is still counted as one flat pool.

That means a retry or earlier expansion task can consume a call that was needed for a later guaranteed production-site task. The final diagnosis is better than before, but the system still allows this sequence:

1. selector chooses enough guaranteed tasks;
2. scheduler admits enough guaranteed tasks;
3. execution starts;
4. an earlier task retry consumes shared recall budget;
5. a later guaranteed task is skipped as budget-limited.

This is not a selection defect anymore. It is a missing external-call reservation rule.

Intended Pipeline Behavior

The benchmark smoke pipeline should treat guaranteed expansion tasks as having a reserved first OpenRouter recall-expansion call.

The intended order is:

1. Build expansion targets.
2. Select guaranteed variants first.
3. Scheduler admits the guaranteed work.
4. Scheduler registers the first-call reservation for every accepted guaranteed recall-expansion task.

- 5. Execution runs guaranteed tasks first.
- 6. Retries and optional tasks may use only the remaining headroom.
- 7. If headroom is exhausted, optional/retry work is limited before it can steal another guaranteed task's first call.

Roles Changed

Role	Change
RadarSearchExpansionService	Still owns target and variant generation; no provider calls.
RadarWorkScheduler	Registers guaranteed recall-expansion first-call reservations during admission.
RadarExternalCallBudget	Protects first provider call per guaranteed task from retries and optional work.
staged execution	Executes the accepted work portfolio and records clearer budget reasons.
benchmark report	Shows selected, admitted, reserved, executed, and skipped lane counts.

Context Passed Between Roles

The scheduler receives selected expansion variants with `schedule_role`.

For every accepted guaranteed expansion item it passes to the external budget:

- `task_id`;
- `reserve_key`;
- `lane`;
- `target_id`;
- `target_type`;
- `schedule_role=guaranteed`.

The budget records:

- protected task ids;
- which protected tasks are guaranteed;
- which guaranteed tasks have spent their first call;
- remaining guaranteed first-call reserve.

Source, Budget, And Checkpoint Semantics

This slice does not change source policy or scoring.

It changes budget semantics:

- the first OpenRouter call of an accepted guaranteed expansion task is protected;
- retries cannot consume the last call needed by another unexecuted guaranteed task;
- optional expansion cannot consume guaranteed first-call reserve;
- `benchmark_smoke` gets small operational headroom above the minimum five guaranteed expansion calls.

New or clarified reasons:

- `guaranteed_external_reservation_insufficient`;
- `guaranteed_external_reservation_protected`;
- `optional_work_budget_limited`;

- openrouter_recall_expansion_budget_limited.

Dossier, Trace, And Evaluation Visibility

The dossier/report should show:

- guaranteed external call reservation count;
- guaranteed external calls used;
- guaranteed external calls remaining;
- failures caused by lack of recall-expansion headroom.

Evaluation does not change metric definitions. It should simply receive a run where the guaranteed production-site lane was actually attempted, or a precise pre-provider reason why this was impossible.

Diagram

Test Plan

- Unit tests for the external budget:
 - protected retry cannot steal the first call reserved for another guaranteed task;
 - guaranteed first call succeeds after retry headroom is blocked;
 - insufficient recall-expansion limit is reported before execution.
- Scheduler tests:
 - accepted guaranteed work registers protected first-call reservations;
 - optional work remains admitted only when headroom exists.
- Benchmark profile tests:
 - benchmark_smoke has recall-expansion headroom above the five-call minimum.
- Dossier/report tests:
 - reservation metadata is visible and contains no secrets.

Acceptance Criteria

- Docker benchmark_smoke reaches a terminal state.
- Selection remains satisfied.
- Scheduler admission remains satisfied.
- External recall-expansion budget protects first calls for:
 - at least one holding/group task;
 - at least two legal/subsidiary tasks;
 - at least two production-site/branch tasks.
- If the second production-site task is still not executed, the reason is not ambiguous shared-budget loss.

Out Of Scope

- No UI change.
- No database migration.

- No new provider adapter.
- No scoring change.
- No SIBUR-specific production hardcode.
- No benchmark quality claim.